J. COMPUTER APPLICATIONS IN ARCHAEOLOGY · 8(1) · 2025

Finding Hillforts at Scale

A deep-learning + LiDAR search for archaeological sites across 180,000 km² of Europe

Landauer · Maddison · Fontana · Posluschny • England · Hesse · Molise

01 — THE PROBLEM

LiDAR outpaces the humans meant to read it

266 GB

LiDAR for England alone — about **1M** patches to scan

<0.05%

of British land is hillfort — a **needle in a haystack**

180,000 km²

searched across **three** European landscapes

> Of **291** reviewed detection projects (Vinci et al. 2024), only **11** exceeded 10,000 km² — and just **5** of those used AI. This is among the first to span three distinct zones with one workflow.

02 — DATASET CREATION

Classification, not segmentation

The Atlas gives centre points but no shapes — so each site becomes a labelled image vignette, and the model predicts *"is there a hillfort here?"*

4,000+	records in the *Atlas of Hillforts of Britain & Ireland*
579	usable positives after pruning destroyed / built-over sites
20,000	random land patches as negatives
raw DEM	min/max normalized — **no hillshade or LRM**

768 × 768 m DEM patch · 1 m resolution

03 — ARCHITECTURE

A ConvNeXt classifier, lightly tuned

BACKBONE	SCHEDULE	AUGMENTATION	LOSS
ConvNeXt	**50 epochs**	**Flip · crop**	**Weighted LS-CE**
ImageNet-pretrained; Fast.ai + PyTorch	fine-tuned from pretrained weights	random H/V flips, resize & crop	label-smoothing cross-entropy for imbalance

`output →` one **confidence score 0–100%** per vignette: the likelihood that hillfort-like features are present.

04 — LABEL REFINEMENT

Pseudo-labelling prunes the noise

Run the trained model back over its own data and drop the contradictions:

`drop` hillforts scoring **< 85%**

`drop` non-hillforts scoring **> 5%**

`retrain` 12 epochs on the cleaned set

BEFORE		AFTER
579 forts 20,000 non-forts	→	505 forts 15,156 non-forts

+4% accuracy from the cleaned retraining pass

05 — INFERENCE & EVALUATION

Sliding the window, scoring the hits

SCAN

384 m	grid step — half-overlapping vignettes scored better
> 90%	confidence = positive detection
cluster	adjacent hits merge into one candidate site

SCORE

TP	cluster within 500 m of a known fort
FP	detection with no fort nearby
FN	known fort the model missed

F1, not accuracy — hillforts are so rare that accuracy would be misleadingly high. True negatives are the entire landscape.

06 — VALIDATION · TABLE 1

High recall, low precision, F1 ≈ 38%

TEST REGION (~2,000 KM²)	KNOWN	PRED.	TP	FP	FN	PREC.	RECALL	F1
Northumberland NP	71	182	49	133	22	0.27	0.69	39%
Northampton (Midlands)	9	15	5	10	4	0.33	0.56	42%
Taunton (Cornwall)	32	108	26	82	6	0.24	0.81	37%
Total	**112**	**305**	**80**	**225**	**32**	**0.26**	**0.71**	**38%**

305 predictions to check instead of **6,000 km²** to eyeball — recall stays high, precision is the cost.

07 — SCALING

All of England in under five hours

~1M	RTX 2060	< 5 hrs	36%
LiDAR patches across 130,000 km^2	single mid-range GPU; multiprocessed I/O	to process the entire country	full-country F1 — matches the test regions

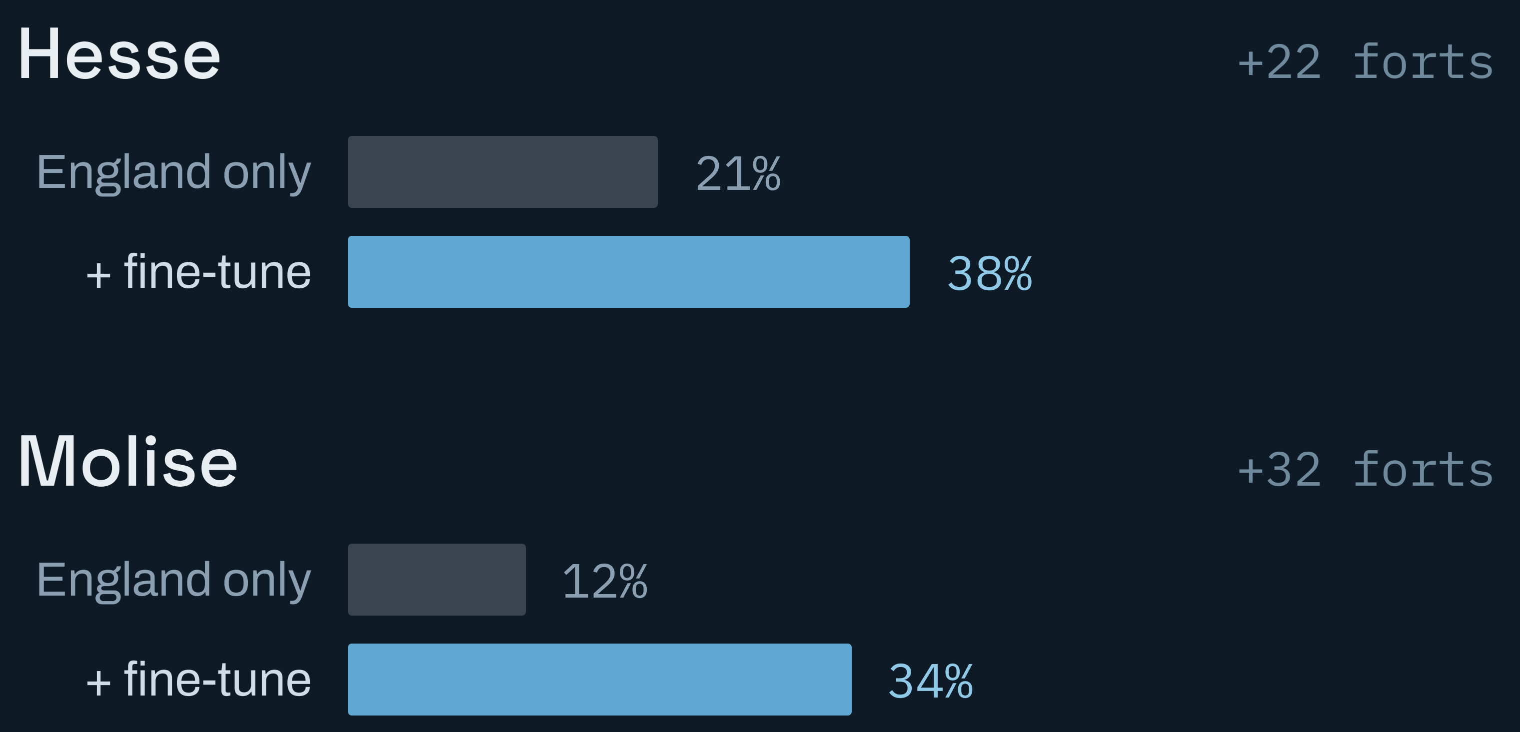
> Consistency between the held-out test areas and the full national run shows the pipeline generalizes within England without re-tuning.

08 — TRANSFER · TABLE 2

Beating model drift cheaply

The English model collapsed abroad — different sensors, terrain, and elevation statistics.

Fine-tuning on **3,000 random local patches** + just **22–32 known forts** (12 epochs) roughly doubled F1.



09 — CASE STUDIES · TABLES 4–5

Real finds in all three regions

England

CHILTERNS + NORTH

1 new

previously unrecognized hillfort, independently confirmed — plus enclosures recorded in the HER but absent from the Atlas.

Hesse

169 CANDIDATES >90%

Forts recovered	12
Possible hillforts	4
False positives	148
Triaged in	~2 hrs

Molise

TOP 100 >0.94

True hillforts	15
Hilltop villages	10
Field systems	10
Terrain features	65

False positives repeat across regions — circular features encircling hills: ring roads, terrace walls, forest tracks.

AI triages; experts confirm

AI CONFIDENCE	PASS 1 N	TIME	PASS 2 N	TIME
90–100%	89	00:17	24	01:37
85–90%	168	00:31	29	01:27
80–85%	235	00:18	11	00:23
75–80%	285	00:07	8	00:05
70–75%	377	00:13	14	00:49
66–70%	346	00:15	9	00:30
Totals	**1,500**	**01:41**	**95**	**04:51**

~4 s

per site, first-pass triage

~3 min

per candidate, deep review

Candidates surfaced even at **66–70%** — a hard cutoff would miss real sites.

11 — TAKEAWAYS

Scalable, transferable — and still human-checked

CONTRIBUTIONS

- → Detects complex, irregular structures — not just simple shapes
- → 180,000 km² — the largest automated archaeological search to date
- → One model transfers across Britain, Germany and Italy

OPEN CHALLENGES

- F1 of 34–38% leaves real headroom for precision
- Maquis vegetation and terraced hills defeat the LiDAR itself
- Next: more regions, faster review, a shared benchmark